On my honor, I have never given nor received any unauthorised and/or inappropriate assistance for this exam. The work done on this exam is totally my own. I understand by the school code that violation of these principles will lead to a zero grade and is subject to a harsh disciplinary issues.

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Q1) Max $z = 4x_1 + 14x_2$ $\quad x_1, x_2, s_1, s_2 \geq 0$

$2x_1 + 7x_2 \leq 21 \rightarrow 2x_1 + 7x_2 + s_1 = 21$
$7x_1 + 2x_2 \leq 21 \quad 7x_1 + 2x_2 + s_2 = 21$
$x_1, x_2 \geq 0$

Basic	z	x_1	x_2	s_1	s_2	sol
z	1	-4	-14	0	0	0
s_1	0	2	7	1	0	21
s_2	0	7	2	0	1	21

MR = 3 ✓
MR = 21/2 = 10,5

$7 - (2 \times \frac{2}{7})$

Basic	z	x_1	x_2	s_1	s_2	sol
z	1	0	0	2	0	42
x_2	0	2/7	1	1/7	0	3
s_2	0	45/7	0	-2/7	1	15

MR = 10,5
MR = 7/3 ✓

Basic	z	x_1	x_2	s_1	s_2	sol
z	1	0	0	2	0	42
x_2	0	0	1	49/315	-14/315	7/3
x_1	0	1	0	-2/45	7/45	7/3

$x_2 = 7/3$
$x_1 = 7/3$
$z = 42$

in iteration 2 and 3, solution is same. There is redundand constraint because it does not effect the solution.

in iteration 2, non basic variable's coefficient is zero

